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ORIGINAL ARTICLE

Rituximab as a sole steroid-sparing agent in generalized myasthenia gravis: Long-term outcomes

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Abstract

Background Rituximab, a B-cell depleting monoclonal antibody, represents an option for the treatment of refractory myasthenia gravis (MG). Its use is more established in muscle-specific tyrosine kinase positive (MuSK+) patients, while its role in managing acetylcholine receptor positive (AChR+), or double seronegative (DSN) patients, remains less clear. This study evaluates the long-term effectiveness and safety of rituximab in MG of various serotypes.

Methods We conducted an open-label study of MG patients receiving rituximab. Adults with generalized refractory MG, either anti-AChR+ or DSN, and anti-MuSK+, refractory or not, who had follow-up > 12 months were selected. Change in quantitative myasthenia gravis (QMG) score at last follow-up, compared with baseline was a primary outcome, as well as factors affecting response to treatment. Secondary outcomes included, long-term safety, the steroid-sparing effect and relapse rates post-rituximab.

Results Thirty patients (16 anti-AChR+, 6 anti-MuSK+, 8 DSN) followed for a mean of 33.3 months were included. Mean scores pre-rituximab compared to last follow-up significantly decreased ($p < 0.001$), from 11 ± 4.1 to 4.3 ± 3.8, and from 1.9 to 0.3 regarding QMG and relapse rate per patient/year, respectively, while in 93.1% a daily steroid dose ≤ 10 mg was achieved. Antibody status was the only factor independently influencing several endpoints. Throughout the study period no crises or deaths occurred.

Conclusion The present study supports that rituximab is an effective and well tolerated treatment for refractory anti-AChR+ and DSN MG patients, while anti-MuSK+ remains the group experiencing the greater benefits.

Keywords Myasthenia gravis · Rituximab · Long-term · Refractory · COVID-19

Introduction

Myasthenia gravis (MG) is an autoimmune disorder of the neuromuscular synapse, with potentially life-threatening complications [1]. The use of immunomodulatory therapies in recent decades has dramatically improved prognosis [2, 3]. Still, a considerable subset of patients (10–30%) remain resistant to conventional therapeutic approaches [4, 5]. Rituximab, a monoclonal antibody directed against B-cell membrane marker CD20, would be a rational biological treatment for MG [2, 3]. To date, its role is more established in MG patients with muscle-specific tyrosine kinase positive (MuSK+) antibodies, who often respond poorly to first-line immunotherapy [3, 6–8]. While there is accumulating evidence for its benefits in non-MuSK serotypes, its place in the treatment of anti-acetylcholine receptor positive (AChR+), and even more so in double seronegative (DSN) patients, remains less clear [2, 3]. Experience so far is based on heterogenous data, often providing conflicting results [9–12].

Among issues regarding rituximab that need further clarification are long-term safety, the magnitude of its effect without concomitant or residual ISs, together with possible prognosticators influencing treatment outcomes [13–15]. The present study reports the long-term outcomes of rituximab in MG patients of various serotype backgrounds, along

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with its safety profile in this sub-group of immunocompromised patients. Data were prospectively collected from a single centre of excellence, with study outcomes specified prior to analysis.

Methods

Study design

We performed an open-label cohort study of patients with MG treated with rituximab. All patients with myasthenia followed at the neuromuscular centre of Patras University Hospital, a tertiary Public University Hospital serving approximately 800,000 people [16], were screened for eligibility through our registry (Fig. 1). For the purpose of this study, all eligible participants were prospectively scheduled for an assessment of their clinical status (irrespective of disease control), which served as last follow-up. Patients' historical records were reviewed for sociodemographic data (age, sex), disease characteristics (disease onset/ duration, antibody status, thymus/ thymectomy), previous MG specific treatments and clinical scores, as well as possible adverse events. Figure 2 portrays the timeline of the present study. Definition of relapses and treatment responses, as well as dosing protocols were based on our institutional policies [16]. No clinical scoring was applied retrospectively.

Definitions

Diagnosis of MG was made based on clinical manifestations of fluctuating muscle weakness and fatigue after exclusion of alternative diagnoses and confirmed by the presence of one or more of the following: detection of AChR or MuSK antibodies; a positive neurophysiological test such as repetitive nerve stimulation (RNS) or single fibre electromyography (SF-EMG); positive neostigmine pharmacological test [17]. Myasthenia was defined as refractory based on the criteria proposed by Mantegazza and Antozzi [4], if any of the following conditions were met: inadequate response ("persistent moderate to severe weakness") to steroids and at least two other ISs; prolonged need of high doses of potentially harmful therapies [7]; requirement of repeated rescue therapies (at least 3 times within a period of 12 months); occurrence of severe or intolerable medication side effects, leading to treatment discontinuation.

Standard clinical scales were employed to define the disease severity and response to treatment [17]. These were: (a) Myasthenia Gravis Foundation of America (MGFA) classification I-V classes; (b) Postintervention Status classified as Complete Stable Remission (CSR), Pharmacological Remission (PR) or Minimal Manifestations (MM), Improved (I), Unchanged (U), Worse-Exacerbation (W/E) or Death from MG (D); (c) Quantitative Myasthenia Gravis (QMG) score (range 0–39), as an objective measure which was available for all patients throughout the study. Sub-scores for individual features of MG were derived by summation of the relevant items from the QMG as follows: ocular (double vision, ptosis), bulbar and neck muscles (facial, swallowing, dysarthria, head lift), limbs (right/left arm/leg outstretch, right/left arm grip), respiratory (vital capacity).

A relapse was defined as a deterioration of MG symptoms appearing after at least a month of remission and lasting no less than 24 h. Two states of relapse severity were recognized: (a) relapse, increase of QMG score by ≥ 3 from 1 or more items and a total QMG score ≥ 6; (b) severe relapse, increase of bulbar and neck QMG score ≥ 3 or respiratory ≥ 1 (irrespectively of total QMG score), or increase of QMG score by ≥ 3 from 1 or more items and QMG total score ≥ 15. Treatment decisions were taken accordingly, i.e., in most cases relapses were dealt with modification of therapeutic regimen, whereas in severe relapses rescue therapy was usually administrated [16].

Relapse rate and rescue therapy rate were defined as the total number of relapses and total number of rescue therapies, respectively, per patient per year. To calculate these rates, at least 2 years of patient monitoring before rituximab initiation was utilized to compare with post-treatment total follow-up time. Regarding dosing frequency, as average cycle time was defined the mean time between cycles per patient calculated for the whole follow-up period and as cycle rate the number of cycles per patient per year, while max cycle time was the longest interval without any repeat cycles per patient. These time intervals reflecting the need to retreatment were used to estimate rituximab dependency. A treatment cycle was considered any of the induction, repeat or maintenance total infusions.

Participants

Patients were eligible for inclusion if: they were adults with generalized refractory MG, either anti-AChR + or DSN, and anti-MuSK + , refractory or not; follow-up more than 12 months since rituximab initiation was completed; surgery was performed at least 12 months before last follow-up in case of thymectomy; any IS agents other than steroids had not been administrated at least 12 months before last follow-up, while rescue therapies such as plasmapheresis (PLEX) or intravenous immunoglobulins (IVIG) were allowed.

This research was performed in accordance with the 1964 Helsinki declaration with its later amendments [18]. All participates gave their written informed consent and the study was approved by the Institutional Ethics Committee of Patras University Hospital (No. of approval 1134 /19–12-2019).

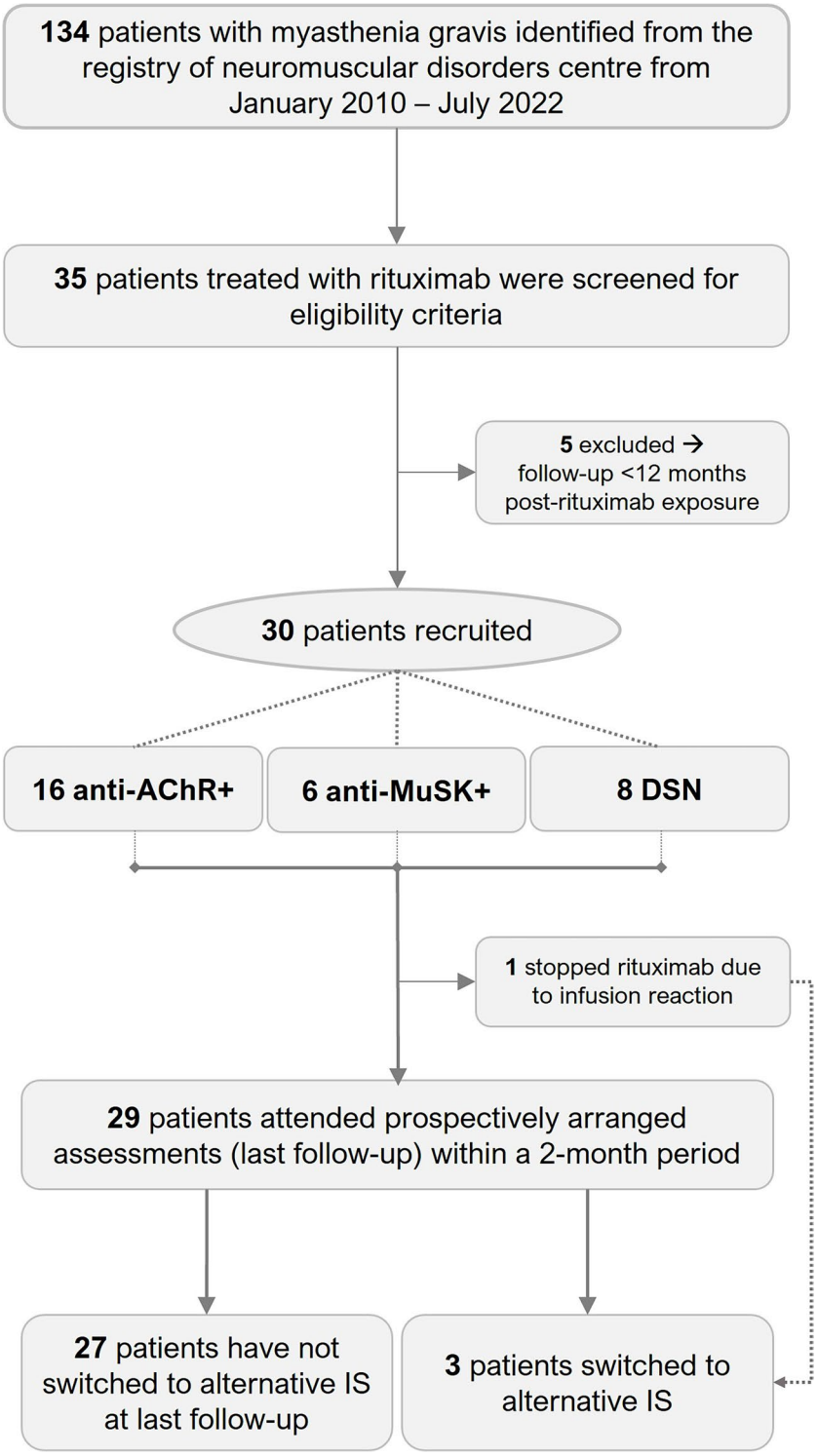

Fig. 1 Study flow chart. Abbreviations. *AChR* acetylcholine receptor; *DSN* double seronegative; *IS* immunosuppressive; *MuSK* muscle-specific tyrosine kinase

Dosing scheme

For anti-AChR + or DSN patients, our institutional protocol included an induction cycle with 4 weekly infusions of 375 mg/m^2, and a default repeat cycle with 2 weekly infusions of 375 mg/m^2 after 6–8 months. Anti-MuSK + patients had induction with 375 mg/m^2 infusions every week for 4 weeks, without a default repeat cycle. Following these standard dosing schemes, maintenance infusions (intervals ≥ 6 months) as two-week 375 mg/m^2 regimes were adjusted by the treating neurologist according to treatment response to avoid

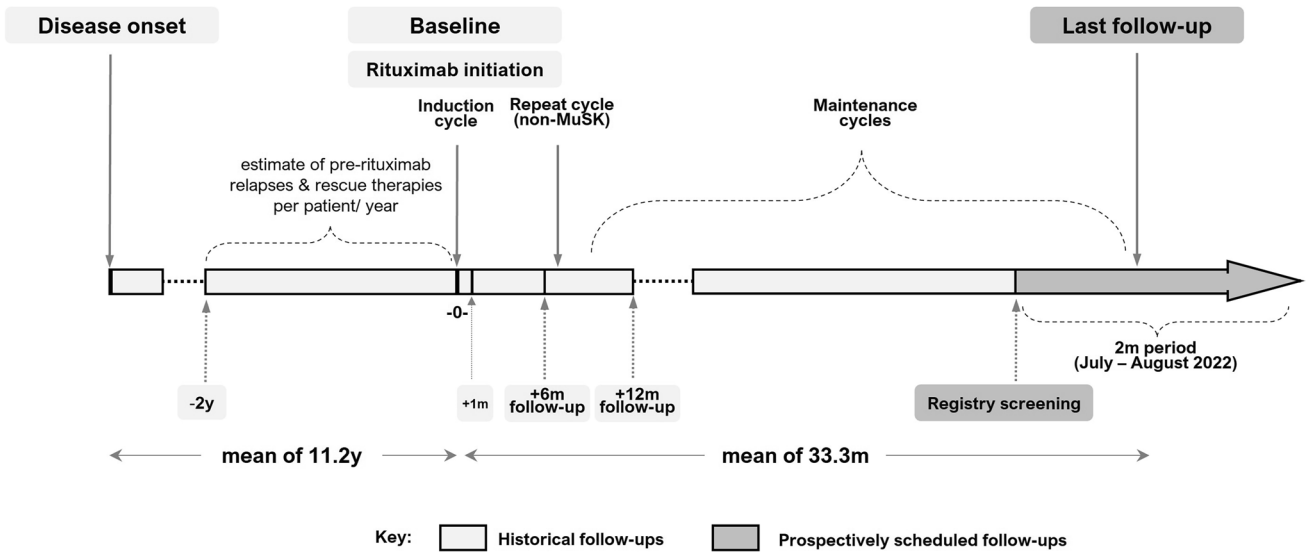

Fig. 2 Study design chart. *Abbreviations. M* months; *MuSK* muscle-specific tyrosine kinase; *y* years

relapses, with close monitoring of possible symptom reappearance [7, 9, 13].

Procedures and assessments

During the last follow-up visit standard clinical scales were used as mentioned above, while the following data were collected by combination of patient interview and review of registry files: medication doses, number and timing of relapses, rescue therapies, and rituximab infusions. Historical data immediately before rituximab initiation (baseline), and 1 month, 6 months and 12 months follow-ups post-rituximab initiation were used for comparisons (Fig. 2).

As for safety, adverse events were systematically collected and ranked (Grade 1 to 5) according to the Common Terminology Criteria for Adverse Events (CTCAE), based on chart review [19]. Additionally, all cases with confirmed severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) by positive rapid and/or polymerase chain reaction (PCR) test were recorded (from 26th February 2020 until last follow-up). Necessity of hospitalization or intensive care unit and correlations with MG relapses were documented.

Outcome measures

Response to treatment was grouped into the following three categories: (a) favourable, reaching CSR, PR or MM stage, and QMG score ≤ 3 from 2 or more items; (b) improvement but not favourable, total QMG score > 3 but QMG score decrease by ≥ 6, or QMG score decrease by ≥ 3 and steroids dose ≤ 10 mg with no relapse the previous 6 months; (c) no improvement or insignificant change if none of the aforementioned conditions were achieved. Categories (a) and (b) were considered as a clinically meaningful response to rituximab.

The prespecified primary endpoints were the change in QMG scores at last follow-up (i.e., prospectively scheduled assessment post-rituximab initiation irrespective of disease control) compared with baseline, and the number of patients achieving a clinically meaningful response at last follow-up (Fig. 2). We also performed a post-hoc analysis to assess the impact of various baseline patient and disease related factors (demographics, serotype, disease duration, steroid dosage, thymus, QMG and MGFA scores) on rituximab response over time. Secondary outcomes were safety and tolerability in the long-term, including COVID-19 related data, the change in QMG scores between baseline and 6, as well as 12 months, and QMG sub-score changes (as a post-hoc exploratory measure), in all follow-up time points. Additional secondary outcomes were the reduction in medication, the relapse and rescue therapy rates, and rituximab dependency as measured by infusions frequency.

Statistical Analysis

Clinical outcomes for continuous values are presented as mean $\pm$ standard deviation (SD), and as numbers (n) and percentages (%) for categorical variables. Differences between groups were analysed by means of paired or unpaired t test for numeric values as applicable, and with chi square (x^2) or fisher exact test for categorical values. Multivariate regression analysis, as guided by univariate analysis outcome, was used to explore the impact of baseline factors to study outcomes. Group differences for time to relapse were assessed through Kaplan–Meier curves and the log-rank test. All reported p values are two-sided at a significance

level of ≤ 0.05. Data were analysed using the SPSS statistical package (IBM SPSS Statistics for Windows, Version 27.0. Armonk, NY).

Results

We identified 134 patients with MG in our registry, out of which 35 have received rituximab. Of those, 30 fulfilled study inclusion criteria and were further evaluated (Fig. 1). Prospectively arranged (last follow-up) visits were completed within a two-month period (between July–August 2022). The mean observation time following rituximab initiation was 33.3 (± 24.7) months. In cases of thymectomy all patients in this cohort had performed surgery at least 2 years before rituximab initiation. Participants had mean disease duration of 11.7 years and mean age at onset of 40.5 years, while the two-thirds were women. The majority of study participants (76.7%) had high disease severity at baseline with MGFA grade III or IV, while 50% had previously tried at least 2 ISs other than steroids. Rituximab was selected as first choice treatment for a total of 8 patients which were either in the anti-MuSK + group (n = 4) or represented severe cases who necessitated repeated rescue therapies (n = 4), requiring a fast-acting agent. Patients' characteristics stratified by serotype are presented in Table 1.

Out of the 30 patients, 29 completed at least 2 treatment cycles, while 1 stopped the treatment due to side effects. At last follow-up, 27 patients (90%) were still on rituximab, while 2 patients failed to respond to rituximab and have switched to another IS (Fig. 1). A meaningful clinical improvement was observed 1 month after completion of the induction cycle of rituximab for 86.2% of patients. Figure 3 portrays levels of response to treatment at the different follow-up points. QMG scores significantly dropped from baseline compared to all follow-up time points (Supplementary Fig. 1), with rituximab providing significant and persistent relief in all symptoms, according to QMG sub-item analysis. Regarding prednisolone, 58.6%, 82.8%, and 93.1% of patients decreased their daily dose to 10 mg or less at 6 months, 12 months, and last follow-up, respectively. Patients needed a mean of 1.7 ± 0.6 treatment cycles per year, while mean cumulative number of treatment cycles per patient was 3.7 ± 1.6 throughout the observational period. Table 2 demonstrates clinical outcomes of rituximab therapy at the different follow-up points.

Multivariate regression was used to measure the impact of patient baseline characteristics such as antibody serotype, age, disease duration, gender, thymus pathology, QMG scores, MGFA and steroid daily dose in clinical response, with respect to QMG scores at last follow-up. These variables significantly explained QMG changes (F = 3.2, p = 0.014, R^2 = 0.644), with antibody serotype being the only factor individually influencing clinical

Table 1 Descriptive characteristics of the study population

	Total	Anti-AChR	Anti-MuSK	DSN
N (%)	30 (100%)	16 (53.3%)	6 (20%)	8 (26.7%)
Gender, N (%)				
Male	10 (33.3%)	5 (31.3%)	3 (50%)	2 (25%)
Female	20 (66.7%)	11 (68.7%)	3 (50%)	6 (75%)
Age in years, mean ± SD				
At disease onset	40.5 ± 13.9	41.4 ± 14.3	39.2 ± 20	39.6 ± 8.3
At last follow-up	52.2 ± 13.8	53.2 ± 14	50 ± 20.1	52 ± 8.5
Disease duration in years, mean ± SD	11.7 ± 7.8	11.6 ± 6.7	10.9 ± 9.3	12.2 ± 9.7
Thymus pathology †, N (%)	18 (60%)	10 (62.5%)	1 (16.7%)	7 (87.5%)
Thymectomy, N (%)	17 (56.7%)	9 (56.3%)	1 (16.7%)	7 (87.5%)
MGFA at baseline, N (%)				
I	1 (3.3%)	0 (0%)	1 (16.7%)	0 (0%)
II	6 (20%)	3 (18.8%)	2 (33.3%)	1 (12.5%)
III	20 (66.7%)	11 (68.7%)	3 (50%)	6 (75%)
IV	3 (10%)	2 (12.5%)	0 (0%)	1 (12.5%)
Previous IS, N (%)				
Steroids alone	8 (26.7%)	1 (6.25%)	4 (66.7)	3 (37.5)
Steroids plus 1 more medication	7 (23.3%)	4 (25%)	2 (33.3)	1 (12.5)
Steroids plus 2 more medications	15 (50%)	11 (68.75%)	0 (0)	4 (50)
Last Follow-up in months, mean ± SD	33.3 ± 24.7	29.5 ± 15.5	57.8 ± 41.4	21.4 ± 8.9

Abbreviations. *AChR* acetylcholine receptor; *DSN* double seronegative; *IS* immunosuppressive; *MGFA* Myasthenia Gravis Foundation of America; *MuSK* muscle-specific tyrosine kinase; *N* number; *SD* standard deviation. †Thymus pathology collectively refers to cases of thymic hyperplasia or thymoma of any class

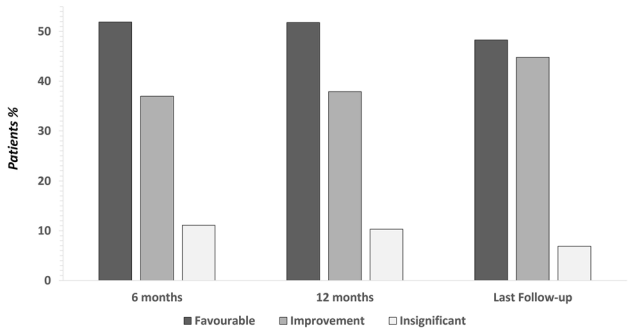

Fig. 3 Proportion of patients in each improvement category at different follow-up points. According to study protocol improvement was classified as: (**a**) favourable, reaching CSR, PR or MM stage, and QMG score ≤ 3 from 2 or more items; (**b**) improvement but not favourable, total QMG score > 3 but QMG score decrease by ≥ 6, or QMG score decrease by ≥ 3 and steroids dose ≤ 10 mg with no relapse the previous 6 months; (**c**) no improvement or insignificant change if none of the aforementioned conditions were achieved. Abbreviations. Complete Stable Remission *(CSR)* Minimal Manifestations *(MM)* Pharmacological Remission *(PR)* Quantitative Myasthenia Gravis *(QMG)*

response (95% CI 0.129–0.487, p = 0.002). Sub-group analysis revealed that, DSN patients had more limited improvement compared to anti-AchR +, while anti-MuSK + had the best outcomes (Supplementary Fig. 1). Regarding rituximab dependency, a similar pattern was observed, with DSN requiring more frequent infusions compared to anti-AChR + patients (Supplementary Fig. 2). Riddance from symptomatic therapy, as another measure of disease control, was also less profound in DSN patients, with mean pyridostigmine reduction at last follow-up 20.2% vs 53.3% in anti-AChR + patients (p = 0.04). Contrary, a longer time to relapse and the lowest rates of treatment infusions were observed in anti-MuSK + group (Supplementary Figs. 2 and 3).

No rebound phenomena, myasthenic crises or deaths occurred during the follow-up period and no patients developed PML or malignancies. One patient experienced severe infusion reaction (laryngeal oedema), after first infusion, and thus treatment was permanently discontinued. This side effect was resolved within hours with appropriate treatment (steroids, adrenaline) with no further sequelae. During COVID-19 pandemic period, 9 patients (30%) with SARS-CoV-2 were identified, with the majority presenting with mild COVID-19 symptoms and 3 needing hospitalization, whereas no deaths occurred [20–22]. Type and severity of AEs possibly related to rituximab are presented in Table 3.

Discussion

We herein report the long-term effect of rituximab in a cohort of patients with refractory or anti-MuSK + MG, following an open-label design with prospective data collection. We noted an overall beneficial effect with 93.1% of our patients improving, while 48.3% reached MM or better. Other significant benefits were the reduction of steroid dose, and the stability of the response as reflected by reduction in relapse rates and the need for rescue therapies. These findings help clarify the role of rituximab as a sole steroid-sparing agent, since prior data mostly report outcomes with the confounding effect of other non-steroid ISs.

Accumulating experience in small case series favours rituximab effectiveness [9, 11, 23, 24]. A double-blind randomized controlled trial (RCT) conducted by Nowak et al. showed no better steroid sparing effects of rituximab vs placebo, in patients with mildly or moderately severe MG although the authors noted a reduction in relapse rate in the rituximab group [12]. Study design issues such as endpoint selection, inclusion of mild patients, with a number of participants already in MM status and low steroid doses at baseline, as well as the use of concomitant ISs, might explain the inability to reveal differences between the two groups, prompting further investigations. In our study, all patients were in a refractory phase, except for anti-MuSK + patients. Our positive findings were similar to those of a previous observational study, also including patients with refractory MG [24]. A recent RCT from the same author group, suggested that a single dose of rituximab as an add-on treatment, increased the probability of newly diagnosed patients to achieve MM status [25]. We therefore argue that rituximab should at least be offered to all MG patients who fulfil the criteria for a refractory phase and when other medications have failed.

Unlike most previous studies where rituximab was given in addition to other ISs, our treatment plan does not include concomitant administration of rituximab with other drugs except for steroids. This practice eliminates the possibility that positive effects of another IS, which could become apparent many months after treatment initiation, might disorientate the results. Another strength of our study is the long monitoring period, which extended not only after rituximab treatment initiation but well before, allowing reliable comparisons between pre- and post-rituximab treatment periods. It was shown that the initial positive response was sustained or further improved throughout the monitoring period. This was evident by QMG scores in regular visits and the reduction of relapse rate in between. The latter finding is important for MG, a disease with unpredicted chronic course and many fluctuations compromising the patients' quality of life [26]. In our study we have incorporated QMG changes and steroid sparing effects to define levels of treatment response as an objective measure of clinical fluctuations. This reflects the long-standing policy that we follow in our department, in order to ensure common practice and communication

Table 2 Effects of rituximab on prespecified primary and secondary outcomes

Score (range)	†Baseline	6-month follow-up	12-month follow-up	Last follow-up (33.3 ± 24.7 months)	6 months vs. baseline	12 months vs baseline	Last follow-up vs. baseline
	Mean ± SD				*Mean difference* *95% CI of the difference* *p value*		
Total QMG (0–39)	11 ± 4.1	4.2 ± 3.6	4.2 ± 3.5	4.3 ± 3.8	*6.5* *5.2–7.9* *< 0.001*	*6.7* *5.3–8.1* *< 0.001*	*6.6* *5.2–7.9* *< 0.001*
Ocular (0–6)	3 ± 1.5	1.5 ± 1.5	1.5 ± 1.4	1.5 ± 1.6	*1.6* *1–2.2* *< 0.001*	*1.6* *0.9–2.1* *< 0.001*	*1.6* *1–2.1* *< 0.001*
Bulbar and neck muscles (0–12)	2.9 ± 1.9	1.0 ± 1.1	0.9 ± 1.1	1 ± 1.2	*1.7* *1.1–2.3 < 0.001*	*1.9* *1.2–2.6* *< 0.001*	*1.8* *1.2–2.4* *< 0.001*
Limbs (0–18)	4.4 ± 2.8	1.6 ± 1.9	1.9 ± 2.1	1.9 ± 2	*2.6* *1.6–3.6* *< 0.001*	*2.4* *1.5–3.4* *< 0.001*	*2.4* *1.5–3.3* *< 0.001*
Respiratory (0–3)	0.7 ± 0.7	0.1 ± 0.2	0 ± 0	0 ± 0	*0.7* *0.4–0.9* *< 0.001*	*0.7* *0.4–1* *< 0.001*	*0.7* *0.4–1* *< 0.001*
Prednisolone, mg/day	27.3 ± 16.3	11.9 ± 4.6	9.3 ± 3.3	7.8 ± 3.3	*15.0* *9.5–20.2* *< 0.001*	*17.6* *11.5–23.6* *< 0.001*	*19.1* *13.1–25* *< 0.001*
Pyridostigmine, mg/day	162 ± 83.6	115.9 ± 84.7	97.2 ± 84	85.9 ± 81.7	*43.5* *21.5–61.3* *< 0.001*	*62.1* *35.9–88.3 < 0.001*	*73.5* *45.9–100.1* *< 0.001*
Relapse rate per patient/year	1.9 ± 0.9	N/A	N/A	0.3 ± 0.4	*N/A*	*N/A*	*1.8* *1.4–2.1* *< 0.001*
Rescue therapy rate per patient/year	1.1 ± 1	N/A	N/A	0.1 ± 0.3	*N/A*	*N/A*	*0.9* *0.6–1.2* *< 0.001*

Abbreviations. *CI* confidence interval; *N/A* not applicable; *QMG* Quantitative Myasthenia Gravis; *SD* standard deviation; *VS* versus. †Baseline refers to the timepoint immediately before rituximab initiation

among neurologists. A similar approach has been utilized by other research groups, who also set cut-off values of QMG changes ≥ 3 for the definition of clinical relapse [25]. Few studies have evaluated the long-term outcome of rituximab in patients with MuSK antibodies only, or refractory myasthenia with either AChR or MuSK antibodies [7, 27–29]. In accordance with our findings, these showed sustainable clinical benefits in 50% of patients or more, lower relapse rates, as well as a steroid sparing effect. A noticeable point is that compared to other ISs, such as azathioprine or mycophenolate, rituximab had a faster onset of action in most patients, thus minimizing the need for bridge therapies. Hence, it represents an optimal treatment strategy for patients with high QMG scores. Similarly, other authors also reported significant improvement after the first cycle of rituximab [5, 29].

Our patients were a representative sample of different antibody profiles. Antibody-specific myasthenia subtype was associated with response to rituximab therapy. In keeping with previous literature, the best outcomes were seen in anti-MuSK + patients [28]. In addition to the initial good response, during the follow-up period, this was the only sub-group to show a continuous improvement over time, while allowing significant reduction of rituximab maintenance cycles. Current consensus recommendations suggest rituximab as a potentially earlier therapeutic option in anti-MuSK + MG [3], as this subgroup has often lower response to first-line symptomatic and IS treatment [6, 7]. Conversely, the DSN sub-group proved to be the most challenging to treat. Despite significant decreases in QMG scores from first rituximab cycle, overall degree of improvement was less compared to the other serotypes. In these patients the interval between cycles could not be extended beyond 6 months without re-appearance of symptoms, indicating treatment dependency. In our previous work we reported that DSN constituted a high percentage of patients with refractory period albeit they eventually benefited by different ISs, as well as rescue treatments [30]. To the best of our knowledge

Table 3 Adverse events during the whole study period

Category	Total N (%†)	§Severe N (%†)	Resolved N (%‡)
Infusion Reaction	**4 (13.3%)**	**1 (3.3%)**	**1 (100%)**
Laryngeal oedema	1 (3.3%)	1 (3.3%)	1 (100%)
Tachycardia	2 (6.7%)	0 (0)	2 (100%)
Hypertension	1 (3.3%)	0 (0)	1 (100%)
Haematologic	**7 (23.3%)**	**0 (0)**	**7 (100%)**
Anaemia	1 (3.3%)	0 (0)	1 (100%)
Lymphopenia	6 (20%)	0 (0)	6 (100%)
Infections	**16 (53.3%)**	**2 (6.7%)**	**16 (100%)**
Respiratory tract	4 (13.3%)	2 (6.7%)	4 (100%)
Urinary Tract	7 (23.3%)	0 (0%)	7 (100%)
Abscess	1 (3.3%)	0 (0%)	1 (100%)
Herpes Zoster	3 (10%)	0 (0%)	3 (100%)
Fungal	1 (3.3%)	0 (0%)	1 (100%)
Covid – 19 related	**11 (33.6)**	**3 (10%)**	**11 (100%)**

† Proportion of total number of patients. ‡ Proportion of events. § Common Terminology Criteria for Adverse Events includes the following ranks: Grade 1: Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated. Grade 2: Moderate; minimal, local or non-invasive intervention indicated; limiting age-appropriate instrumental activities of daily living; Grade 3: Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care activities of daily living; Grade 4: Life-threatening consequences; urgent intervention indicated; Grade 5: Death related to adverse event. Grade > 2 is considered as severe adverse event and in our study, maximum observed severity was of Grade 3. This number represents total number of events as 2 patients were twice detected with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)

there is scant evidence commenting on rituximab effectiveness, specifically for DSN patients. Regarding other potential predictive factors, we found no correlation between age and benefits from rituximab, corresponding with previous data [31]. Our analysis also revealed that positive effects were irrespective of disease duration, suggesting that rituximab remains a valid option even after long disease courses. Considering the high proportion of treatment resistant patients referred to our centre, rituximab proved to be a useful therapeutic tool against difficult cases.

A limitation of the study was the non-double blinded design, which in our view would not be suitable given the severity of symptoms of our patients and the long follow-up period (mean time nearly 3 years) in the absence of concomitant ISs. Anti-musk patients were included collectively, considering that they represent a challenging subgroup of MG patients with frequent bulbar involvement and variable response to first line treatments [3]. However, the permission of refractory, as well as non-refractory anti-MuSK patients, might have influenced the results. Study design has not allowed the inclusion of any naïve rituximab patients, as well. To eliminate bias as possible, all the eligible participants attended last follow-up visit at a random time point in their disease course, irrespectively of disease status, success of rituximab therapy, and whether they were still on the drug or not. Moreover, we used a standard dose for all participants for at least the first year, and therefore efficacy of different dosages is not addressed in the present study. Other groups have suggested the use of much lower dosages with promising results, although rituximab was mostly used as an add-on therapy [24, 25, 32]. Appropriate timing for repeat rituximab infusions to avoid relapses, while minimizing exposure to unnecessary immunosuppression load remains an open question [15, 33]. In the current study, decisions for rituximab retreatment were guided by reappearance of myasthenic symptoms since an established biomarker was not available throughout the study. Measuring B-cell resurgence has been considered as a promising indicator that could allow a tailored treatment approach. However, in the current literature the optimum biomarker for predicting the necessity of rituximab retreatment, such as subset and threshold of B-cell subpopulations or any combination of these, are still under discussion [15, 33–35].

The advent of COVID-19 has posed an additional challenge in the management of MG patients, a particularly susceptible population due to their respiratory muscle involvement and/or chronic immunosuppression [36]. The detailed recording of side effects including the 2-year of COVID-19 pandemic showed no life-threatening events, although infections were unavoidable, in agreement to previous literature [37, 38]. We could point out that rituximab substantially reduces corticosteroids, bulbar and respiratory symptoms which consistently arise as modifiable predictors of worse outcomes in MG patients with COVID-19 [37, 38]. Overall, 3 (10%) severe AEs occurred, all resolved without permanent sequalae. No PML cases and no malignancies were documented in the long-term, with some patients having infusions up to 10 years, corroborating recent evidence that rituximab might be safely regarded as an earlier treatment option [24, 25].

In conclusion, the choice of IS medication as 2nd or 3rd options for anti-AChR + or DSN patients, with moderate to severe symptoms, remains a difficult decision to make [39]. Our results support that rituximab represents a possible third-line candidate, since it offers a fast and long-lasting beneficial effect with a good safety profile. Particularly for MuSK myasthenia, the administration of rituximab seems to eliminate the refractoriness of this sub-group. Unavoidably, its place regarding fast and long-standing responsiveness, safety profile, dosing conveniency, and cost-effectiveness, compared to complement and neonatal Fc inhibitors will have to be determined through future research [40]. Until data for these emerging therapies are accumulated, we point out rituximab as one reliable and safe choice with cost-effective profile, among third line options.

Abbreviations AChR: Acetylcholine receptor; CTCAE: Common Terminology Criteria for Adverse Events; CSR: Complete Stable Remission; CI: Confidence interval; COVID-19: Coronavirus disease 2019; DSN: Double seronegative; IS: Immunosuppressive; IVIG: Intravenous immunoglobulin; MM: Minimal Manifestations; MuSK: Muscle-specific tyrosine kinase; MG: Myasthenia gravis; MGFA: Myasthenia Gravis Foundation of America; N/A: Not applicable; N: Number; PR: Pharmacological Remission; PLEX: Plasmapheresis; PML: Progressive multifocal leukoencephalopathy; QMG: Quantitative Myasthenia Gravis; RCT: Randomized Controlled Trial; SARS-CoV-2: Severe acute respiratory syndrome coronavirus 2; SD: Standard deviation; VS: Versus

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

Conflicts of Interest The authors have no conflicts to disclose related to this manuscript.

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